

Maksimum punte: <i>Maximum marks:</i>	16	Volpunte: <i>Full marks:</i>	15
Duur van vraestel: <i>Duration of paper:</i>	30 min + 10 min to complete OCR form 30 min + 10 min om OKH vorm te voltooi	Oopboek / toeboek: <i>Open / closed book:</i>	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>	8 + OCR		

Important

- Die eksamenregulasies van die Universiteit van Pretoria geld
The examination regulations of the University of Pretoria apply.
- **All questions must be answered on the Optical Character Recognition (OCR) form.**
Alle vrae moet op die Optiese Karakter Herkenning (OKH) vorm ingevul word.
- Read the instructions on the OCR form carefully.
Neem noukeurig kennis van die instruksies op die OKH vorm
- Answers must include units!
Antwoorde moet eenhede insluit!
- **Final answers must be written as real numbers and not as fractions.**
Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.

Dosent(c) / *Lecturer(s):* D le Roux, JP de Villiers, JP Jacobs

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (0-5 min)

Do step 1 in the first 30 minutes of the test. (5-35 min)

Do step 2 within the following 10 minutes. (35-45 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 30 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (30 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN122C1**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

Examples

01	-	1	.	4	5					μ	A						
02	3	4	0							V							
03	1	5	.	2						k							
04	2	0								Ω							

Questions 01-02 [2]		Vrae 01-02 [2]	
A Raspberry Pi 3 operates on a battery rated at 3.4 V and can store 20 000 milliwhatt-hours of energy.		'n Raspberry Pi 3 funksioneer met 'n battery gegradeer teen 3.4 V en kan 20 000 milliwhatt-uur energie stoor.	
01	What is the average current the battery can deliver over a period of 5 hours if it is fully discharged at the end of that period?	01	Wat is die gemiddelde stroom wat die battery kan lewer oor 'n tydskuur van 5 ure indien die battery ten volle ontlai is aan die einde van die tydperk?
02	What is the average power delivered in question 01?	02	Wat is die gemiddelde drywing gelever in vraag 01?

01) $W = 20 \text{ Wh}$

$$W = \int_{t_0}^t p \, dt = p \Delta t = W \Delta t \therefore i = \frac{W}{V \Delta t} = \frac{20}{3.4} = 1.333 \text{ A} \rightarrow$$

02) $P = \frac{W}{t} = \frac{20}{5} = 4 \text{ W} \rightarrow$

Question 03 [1]		Vraag 03 [1]	
Consider the charge flowing from point <i>a</i> to point <i>b</i> in a circuit. The graph of the charge $q(t)$ at point <i>b</i> is shown in the Figure 1 below.		Beskou die lading wat vloei vanaf punt <i>a</i> na punt <i>b</i> in 'n bepaalde stroombaan. Die grafiek van die lading $q(t)$ by punt <i>b</i> word getoon in Figuur 1 hieronder.	
03	What is the current from <i>a</i> to <i>b</i> at time $t = 0.25 \text{ s}$?	03	Wat is die stroom vanaf <i>a</i> na <i>b</i> by tyd $t = 0.25 \text{ s}$?

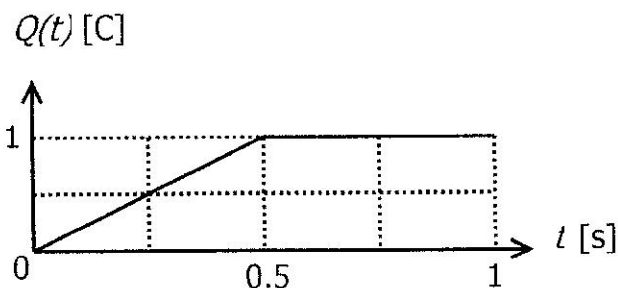


Figure 1

03) $i = \frac{dQ(t)}{dt} = \frac{\Delta Q}{\Delta t} = \frac{1}{0.5} = 2 \text{ A} \rightarrow$

Question 04 [1]		Vraag 04 [1]	
How many independent loops (l) and branches (b) are there in the circuit in Figure 2.		Hoeveel onafhanklike lusse (l) en takke (b) is daar in die stroombaan in Figuur 2.	
05	Number of independent loops l ?	05	Hoeveelheid onafhanklike lusse l ?

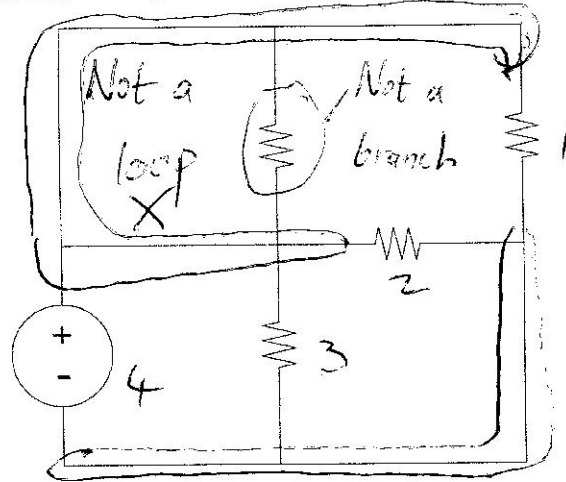


Figure 2

05) $l = 3 \rightarrow n = 2$

$b = 4$

Test

$$\begin{aligned}
 b &= l + n - 1 \\
 &= 3 + 2 - 1 \\
 &= 4 \rightarrow
 \end{aligned}$$

Question 05 [2]		Vraag 05 [2]	
05	Calculate the power P_1 relating to the dependent current source in Figure 3.	05	Bereken die drywing P_1 met betrekking tot die afhanklike stroombron in Figuur 3.

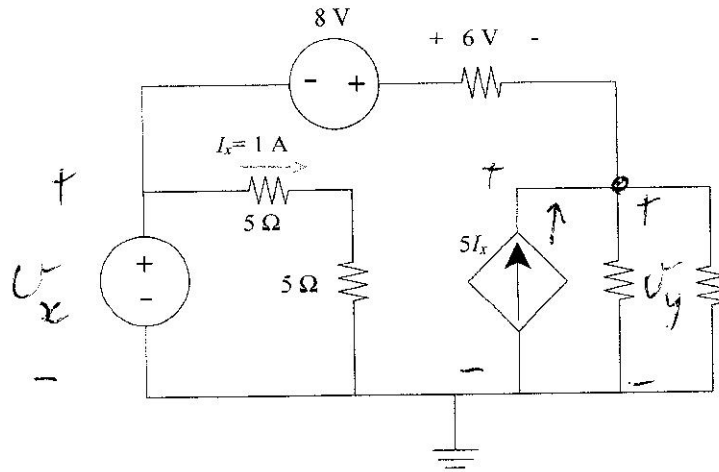


Figure 3

$$V_x = 1(5+5) = 10 \text{ V}$$

KVL

$$-V_x - 8 + 6 + V_y = 0$$

$$-10 - 8 + 6 + V_y = 0$$

$$V_y = 12 \text{ V}$$

$$P_1 = -V_y I$$

$$= -12(5)$$

$$= -60 \text{ W} \rightarrow$$

Questions 06-07 [4]		Vrae 06-07 [4]	
Consider the circuit in Figure 4.		Beskou die stroombaan in Figuur 4.	
06	Calculate the voltage v_2 .	06	Bereken die spanning v_2 .
07	Calculate the current i_2 .	07	Bereken die stroom i_2 .

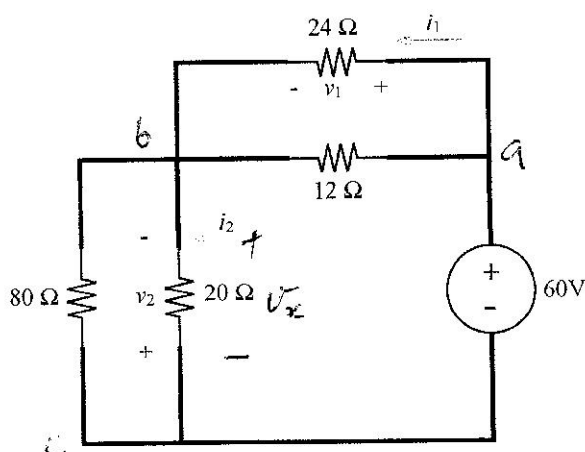


Figure 4

$$06) \quad \frac{1}{R_{ab}} = \frac{1}{24} + \frac{1}{12} = \frac{3}{24} = \frac{1}{8} \quad \therefore R_{ab} = 8\Omega$$

$$\frac{1}{R_{bc}} = \frac{1}{80} + \frac{1}{20} = \frac{5}{80} = \frac{1}{16} \quad \therefore R_{bc} = 16\Omega$$

$$\text{Voltage divider: } v_x = 60 \frac{R_{bc}}{R_{ab} + R_{bc}} = \frac{16}{24} 60 = 40V$$

$$v_2 = -v_x = -40V \rightarrow$$

$$07) \quad v_1 + v_x = 60$$

$$v_1 + 40 = 60 \quad \therefore v_1 = 20V$$

$$i_1 = \frac{20}{24} = 0.8333 \text{ A}$$

Question 08 [2]		Vraag 08 [2]	
Consider the circuit in Figure 5.		Beskou die stroombaan in Figuur 5.	
08	Calculate the equivalent conductance between nodes <i>a</i> and <i>b</i> .	08	Bereken die ekwivalente konduktansie tussen nodes <i>a</i> en <i>b</i> .

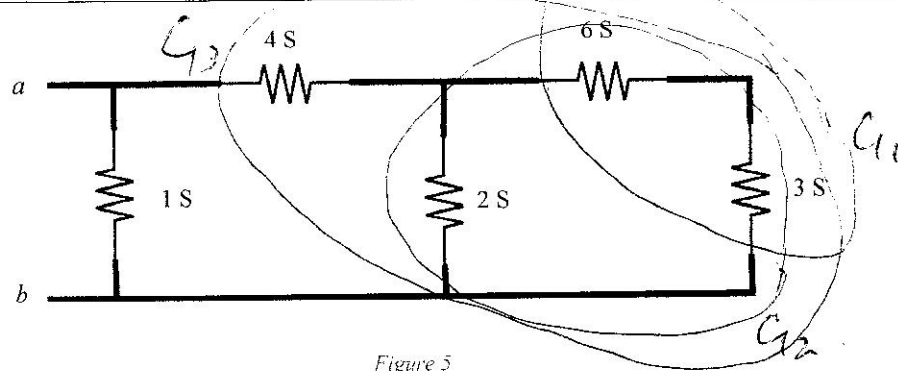


Figure 5

$$08) \quad \frac{1}{G_1} = \frac{1}{6} + \frac{1}{3} = \frac{3}{6} = \frac{1}{2} \quad \therefore G_1 = 2S$$

$$G_2 = G_1 + 2S = 4S$$

$$\frac{1}{G_3} = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2} \quad \therefore G_3 = 2S$$

$$G_{ab} = G_3 + 1S = 3S \rightarrow$$

Question 09-12 [4]				Vraag 09-12 [4]			
Consider Figure 6. Use nodal analysis to set up two equations of the following form:				Beskou Figuur 6. Gebruik nodale analise om die volgende twee vergelykings op te stel:			
$aV_1 + bV_2 = 10$ (node 1)				$aV_1 + bV_2 = 10$ (nodus 1)			
$cV_1 + dV_2 = 20$ (node 2)				$cV_1 + dV_2 = 20$ (nodus 2)			
Determine the coefficients a , b , c , and d .				Bepaal die koëffisiënte a , b , c en d .			
09	a	10	b	09	a	10	b
11	c	12	d	11	c	12	d

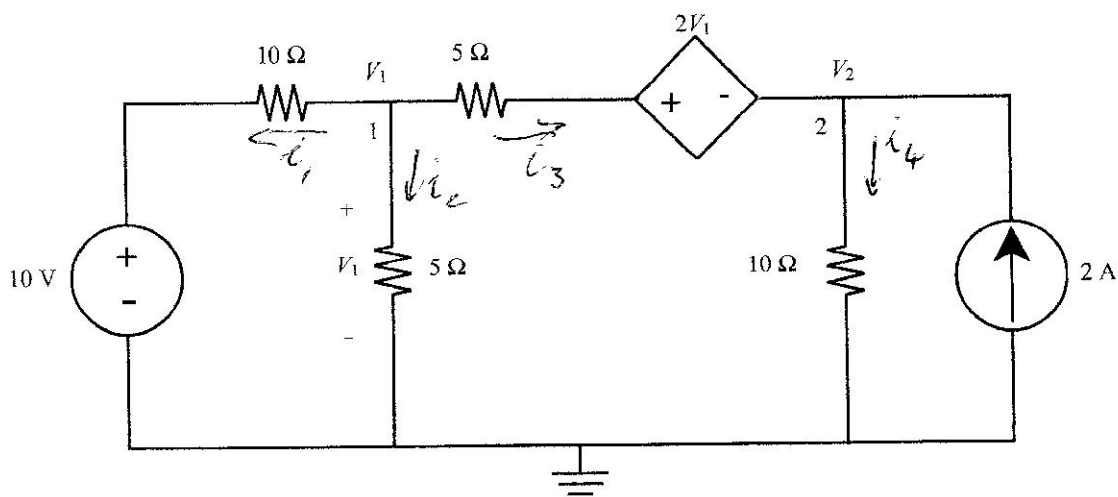


Figure 6

Node 1 : $i_1 + i_2 + i_3 = 0$

$$\frac{V_1 - 10}{10} + \frac{V_1}{5} + \frac{V_1 - (V_2 + 2V_1)}{5} = 0$$

$$V_1 - 10 + 2V_1 + 2V_1 - 2V_2 - 4V_1 = 0$$

$$V_1 - 2V_2 = 10$$

$$a = 1 \quad b = -2$$

Node 2 $i_3 + 2 = i_4$

$$\frac{V_1 - (V_2 + 2V_1)}{5} + 2 = \frac{V_2}{10}$$

$$2V_1 - 2V_2 - 4V_1 + 20 = V_2$$

$$-2V_1 - 3V_2 = -20$$

$$2V_1 + 3V_2 = 20$$

$$\left| \begin{array}{l} c = 2 \\ d = 3 \end{array} \right.$$

PRACTICE PAGE / OEFENBLAD

Assessment ID

2	0	1	8	E	B	N	1	2	2	C	1		
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Answer

Unit

01	$i = 1.333$	A
02	$p = 4$	W
03	$i(0.25s) = 2$	A
04	$\#l = 3$	No units
05	$P_1 = -60$	W
06	$v_2 = -40$	V
07	$i_t = 0.8333$	A
08	$G_{ab} = 3$	S
09	$a = 1$	No units
10	$b = -2$	
11	$c = -6$	
12	$d = 3$	
13		
14		
15		